

SRS

DOCUMENT

EMPLOYEE MANAGEMENT SYSTEM

NAME :- MOHAMMED FAROOQ.S

SUBJECT :-SOFTWARE ENGINEERING

SECTION :-BCA 5TH SEM

TOPIC :-EMPLOYEE MANAGEMENT
SYSTEM

SUBMITTED TO :-SUJANA MAMDAM

1. Introduction

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the **Employee Management System (EMS)**. The EMS is intended to manage employee records, attendance, leave management, payroll, and generate reports, providing an efficient and streamlined solution for human resources management.

2. Overall Description

2.1 Purpose

The Employee Management System (EMS) will automate and facilitate the management of employee records, attendance, payroll, leave requests, and other HR processes. It will be used by various users, including administrators, managers, and employees, to perform specific tasks related to employee management.

2.2 Scope

- The system will support employee data management (personal and professional).
- It will track attendance, process leave requests, and manage payroll.
- The system will generate reports for employees, managers, and administrators.
- It will allow users to log in based on roles (Admin, Manager, Employee).

2.3 Intended Audience

- **System Administrators:** Responsible for managing employee records, system settings, and overseeing operations.
- **Managers:** Responsible for reviewing employee records, managing leaves, and generating reports.
- **Employees:** Can view personal data, request leaves, and access payroll information.

2.4 Product Overview

The EMS is a web-based application designed to be intuitive, user-friendly, and accessible via any modern browser. It will include role-based access controls, a secure login system, and automated payroll processing.

3. User Requirements Specification (URS)

3.1 User Profiles

- **Administrator:**
 - Full access to all functionalities (employee management, payroll, reports, system settings).
 - Can manage employee records, generate reports, and configure system settings.
- **Manager:**
 - Can approve or reject leave requests.
 - Can generate departmental reports and monitor employee attendance.
 - Cannot add or delete employees, but can view and modify employee data in their department.
- **Employee:**
 - Can view personal data (attendance, leave balance, payroll).
 - Can submit leave requests and track their approval status.
 - Can view and download salary slips.

3.2 Functional Requirements

- **Employee Management:** The system should allow adding, viewing, editing, and deleting employee records (name, department, position, salary).
- **Attendance Management:** Employees should be able to mark their attendance (via manual entry or biometric integration). Managers and Admin should view and modify attendance records if needed.
- **Leave Management:** Employees should be able to request leave, and managers should approve or reject those requests. Leave balances should be updated automatically.

- **Payroll Management:** The system should calculate payroll based on employee attendance and leave data and allow employees to view their payslips.
- **Reporting and Analytics:** The system should provide various reports such as attendance, payroll, and leave balances. Reports should be exportable to PDF or Excel.
- **User Role Management:** The system should support multiple user roles (Admin, Manager, Employee) with different access rights.

3.3 Non-Functional Requirements

- **Performance:** The system should support up to 1,000 concurrent users.
 - **Security:** The system should support encrypted communication (SSL/TLS) and strong password policies for login. Role-based access control should be implemented.
 - **Usability:** The system interface should be simple and intuitive, with help documentation available for users.
-

4. System Requirements Specification (SRS)

4.1 Functional Requirements

- **Employee Data Management:**
 1. Add, update, delete employee records.
 2. View personal and professional information (e.g., position, department, salary).
- **Attendance Tracking:**
 1. Employees can mark their daily attendance.
 2. Attendance can be manually edited by the Admin or Manager.
 3. Attendance data should be logged and used for payroll calculations.
- **Leave Request and Approval:**
 1. Employees can submit leave requests for different types of leave (e.g., sick leave, vacation).
 2. Managers can approve or reject leave requests.
 3. The system will automatically update leave balances after approval.
- **Payroll Processing:**

1. Payroll will be automatically calculated based on attendance, leave, and salary details.
 2. Employees can download their salary slips.
- **Reports and Analytics:**
 1. Admin and Managers can generate employee performance, attendance, payroll, and leave reports.
 2. Reports can be exported in PDF or Excel formats.

4.2 Non-Functional Requirements

- **Performance:**
 - The system must be able to generate reports in less than 30 seconds.
 - The system should load user dashboards within 5 seconds.
- **Security:**
 - SSL/TLS encryption for all data in transit.
 - Role-based access control to restrict access to sensitive data.
- **Usability:**
 - The system should be easy to navigate, with a responsive design.
 - The user interface should be intuitive, with clear navigation paths for Admin, Manager, and Employee roles.
- **Scalability:**
 - The system should scale easily to handle more employees and users as the organization grows.
 - Cloud infrastructure should be used to support scalability and high availability.

5. Software Requirements

5.1 Software Specifications

- **Frontend:**
 - HTML5, CSS3, JavaScript for basic UI components.
 - Frontend framework like React.js or Vue.js for dynamic user interfaces.

- **Backend:**
 - Node.js with Express.js, Python (Django/Flask), or Java (Spring Boot) for backend API.
- **Database:**
 - Relational Database (e.g., MySQL or PostgreSQL) to store employee, attendance, payroll, and leave data.
- **Authentication and Authorization:**
 - JWT (JSON Web Tokens) or OAuth for secure login and user management.
- **Web Server:**
 - Apache or Nginx for handling requests and serving the application.

5.2 System Constraints

- **Cloud Deployment:**
 - The system should be deployed on a cloud service like AWS, Azure, or Google Cloud for easy scaling.
 - **Third-party Integrations:**
 - Biometric attendance systems (optional).
 - Email service for notifications (e.g., Gmail API, SendGrid).
-

6. Hardware Requirements

6.1 Client-Side Requirements

- Any modern web browser (Chrome, Firefox, Safari, Edge).
- Minimum screen resolution of 1366x768 pixels.
- Stable internet connection for accessing the web-based system.

6.2 Server-Side Requirements

- **Web Server:**
 - 2-4 CPU cores.
 - 8 GB RAM minimum.
 - 100 GB SSD storage for data (expandable).

- Network bandwidth: 1 Gbps.
 - **Database Server:**
 - 4-8 CPU cores.
 - 16 GB RAM minimum.
 - 500 GB SSD storage.
 - Database software (MySQL or PostgreSQL).
 - **Backup Server (optional):**
 - For periodic backups of the employee and payroll data.
-

7. System Design and Architecture

7.1 Architecture Overview

- The system will follow a three-tier architecture:
 1. **Frontend (Presentation Layer):** Handles user interaction.
 2. **Backend (Business Logic Layer):** Processes data and handles business logic.
 3. **Database Layer:** Stores employee data, attendance, payroll, etc.

7.2 Deployment Diagram

- The system will be deployed on a cloud-based infrastructure, with a load balancer distributing traffic across multiple servers. The database will be hosted on a separate cloud instance to ensure high availability.
-

8. Appendix

8.1 Glossary

- **EMS:** Employee Management System.
- **Admin:** Administrator with full access to all system features.
- **Manager:** User responsible for approving leave requests and monitoring employee performance.
- **Employee:** User who interacts with their own data, including attendance, leave, and payroll.

-----*****-----

*******THANK YOU*******

-----*****-----